

Technology Stack

Date	08 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build the solution. Fill out the table below to detail the specific technologies chosen for each component of the architecture and a brief justification for the choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Jinja2 templating)	Provides a lightweight, responsive interface that loads quickly for end-users without extra dependencies.
2	Backend / Server-Side	Python (Flask)	Excellent integration with the PyTorch model for real-time inference and serving the HTML dashboard.
3	Database / Data Storage	cuDF (RAPIDS GPU DataFrame) / Pandas, CSV-based	GPU-accelerated handling of large, structured, multi-year socio-economic datasets.
4	Cloud / Hosting / Deployment	LocalTunnel (via npx), Kaggle Notebook GPU runtime (Tesla T4)	Enables free, on-demand public access to the Flask app directly from the training environment.
5	Version Control & CI/CD	GitHub	Essential for team collaboration, version tracking, and documenting each project phase.
6	Third-Party APIs / Other Tools	PyTorch, scikit-learn, Matplotlib, Seaborn	Used for model development (hybrid neural network), evaluation metrics, and visualization of results.